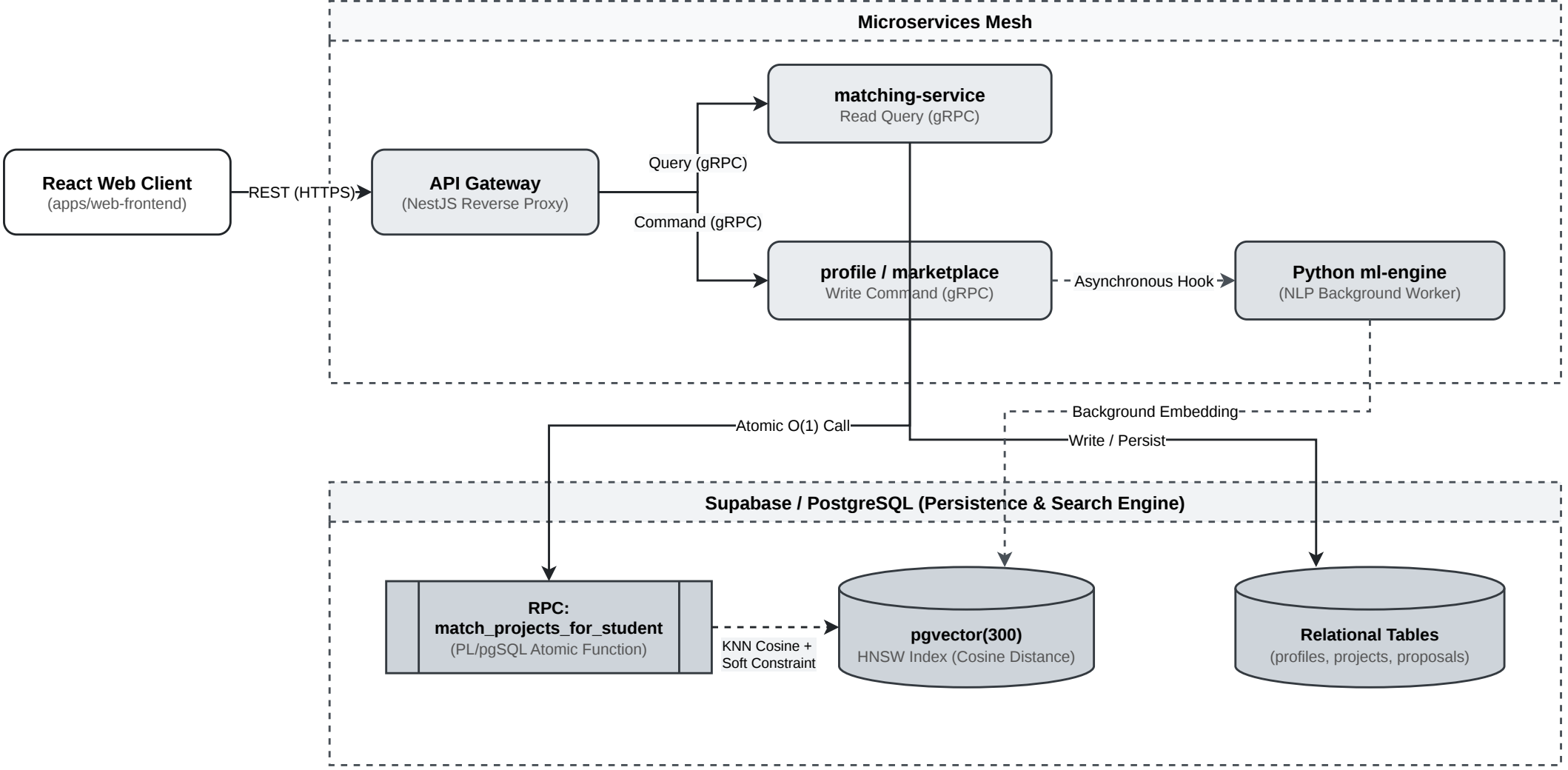
Microservices Mesh
React Web Client
(apps/web-frontend)
REST (HTTPS)
API Gateway
(NestJS Reverse Proxy)
Query (gRPC)
Command (gRPC)
matching-service
Read Query (gRPC)
profile / marketplace
Write Command (gRPC)
Asynchronous Hook
Python ml-engine
(NLP Background Worker)
Background Embedding
Atomic O(1) Call
Write / Persist
Supabase / PostgreSQL (Persistence & Search Engine)
RPC:
match_projects_for_student
(PL/pgSQL Atomic Function)
KNN Cosine +
Soft Constraint
pgvector(300)
HNSW Index (Cosine Distance)
Relational Tables
(profiles, projects, proposals)